Measure Theory
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Introduction: From Riemann to Lebesgue

Measure theory provides a systematic way to assign numerical values to sets and functions. Specifically, we assign a **measure** to a measurable set and an **integral value** to an integrable function. While Riemann integration is a staple of introductory calculus, Lebesgue integration theory represents a significant generalization and completion of those ideas. Lebesgue's framework allows us to assign numbers to a much broader collection of sets and functions than is possible within the Riemann system.

The Shift from Finite to Countable Operations

The fundamental difference between these two theories can be summarized by their relationship with infinity:

- **Riemann Integration** is centered on **finiteness**. It relies on approximations of a finite nature, such as summing the areas of finitely many rectangles to find the area of a bounded subset of $\mathbb{R}^2$. This theory is robust when dealing with finite operations on sets $A_1, \ldots, A_n$ or functions $f_1, \ldots, f_n$.
- **Lebesgue Integration** is centered on **countable infiniteness**. Riemann integration often fails to behave consistently when faced with operations of a countably infinite nature, such as the union $\bigcup_{n=1}^{\infty} A_n$, the sum $\sum_{n=1}^{\infty} f_n$, or the limit $\lim_{n\to\infty} f_n$. Lebesgue integration was developed to rectify these disadvantages.

Methodological and Structural Differences

Beyond their handling of infinity, the two theories differ in their execution and scope:

- **Partitioning:** In Riemann's theory, we partition the **domain** of the function, whereas in Lebesgue's theory, we partition the **range**.
- **Generalization:** Riemann's theory is largely restricted to Euclidean space. Conversely, the principles of Lebesgue's theory are applicable to more general spaces, leading to the development of **abstract measure theory**. This abstract approach is highly versatile and intersects with numerous branches of mathematics.

I suggest all students read these two series of notes by **Prof. A. K. Nandakumaran**, Department of Mathematics, IISc Bangalore. These are fantastic articles that provide excellent motivation for studying Measure Theory:

- Meaning of Integration 1
- Meaning of Integration 2

The Vitali Construction: Limits of Generalizing Volume

In this section, we attempt to generalize the concept of volume (specifically length in $\mathbb{R}$) by searching for a set function μ that assigns a "size" to subsets of $\mathbb{R}$. We want this function μ to satisfy certain natural properties:

1. **Normalization:** The length of the unit interval is one, i.e., $\mu([0,1]) = 1$.
2. **Countable Additivity:** If $E_1, E_2, \ldots$ are disjoint subsets, then the measure of their union is the sum of their individual measures: $\mu(\bigcup_{j=1}^{\infty} E_j) = \sum_{j=1}^{\infty} \mu(E_j)$.
3. **Translation Invariance:** If a set E is shifted by a distance r to form a new set $F = E + r$, then $\mu(F) = \mu(E)$.

The following construction, known as the **Vitali Set**, demonstrates that these three assumptions are mutually inconsistent. It proves that there exist sets for which a consistent "length" cannot be defined.

Step-by-Step Construction of the Vitali Set

We work within the interval $[0,1]$ in $\mathbb{R}$.

Step 1: Define an Equivalence Relation We define a relation $\sim$ on $[0,1]$. For any $a, b \in [0,1]$, we say $a \sim b$ if and only if their difference is a rational number:

$$a \sim b \iff a - b \in \mathbb{Q}$$

This partitions $[0,1]$ into disjoint equivalence classes, where each class contains all points that differ from each other by only a rational value.

Step 2: Form the Vitali Set Y Using the **Axiom of Choice**, we construct a set $Y \subset [0,1]$ by picking exactly one element from each equivalence class. This set Y is our "Vitali Set."

Step 3: Analyze Rational Translations Let $Q = \mathbb{Q} \cap [-1,1]$ be the set of rational numbers between -1 and 1. Since $\mathbb{Q}$ is countable, we can list these rationals as $\{r_1, r_2, r_3, \dots\}$. Consider the translated sets:

$$Y_i = Y + r_i = \{y + r_i : y \in Y\}$$

Step 4: Verify Disjointness We claim that if $r_i \neq r_j$, then $Y_i \cap Y_j = \emptyset$. Suppose there was an element in both. Then $y_1 + r_i = y_2 + r_j$ for some $y_1, y_2 \in Y$. This implies $y_1 - y_2 = r_j - r_i$. Since r_j and r_i are rational, their difference is rational, meaning $y_1 \sim y_2$. However, since Y contains only one element from each class, we must have $y_1 = y_2$, which forces $r_i = r_j$. Thus, the translates are disjoint.

Step 5: Containment and Contradiction Note that since $Y \subset [0,1]$ and each $r_i \in [-1,1]$, every translated set Y_i is contained within the interval $[-1,2]$. Therefore:

$$\bigcup_i Y_i \subset [-1,2]$$

By translation invariance, all Y_i must have the same measure: $\mu(Y_i) = \mu(Y) = \delta$. By countable additivity:

$$\mu\left(\bigcup_i Y_i\right) = \sum_{i=1}^{\infty} \mu(Y_i) = \sum_{i=1}^{\infty} \delta$$

- If $\delta = 0$, the total measure is 0. But the union of these translates actually covers the interval $[0,1]$, so the measure must be at least 1. (Contradiction: $0 \geq 1$)
- If $\delta > 0$, the infinite sum $\sum \delta$ diverges to ∞. But the union is contained in $[-1,2]$, so the measure cannot exceed 3. (Contradiction: $\infty \leq 3$)

Thus, no such function μ can exist for all subsets of $\mathbb{R}$.

Step 6: Finite Additivity Failure Suppose $\mu(Y) = \delta$. Since $Y \subset [0,1]$, δ must be some non-negative value.

- If $\delta = 0$, then any countable union of its translates would also have measure 0, which cannot cover $[0,1]$ (as $\mu([0,1]) = 1$).
- If $\delta > 0$, we can find a natural number k such that $k\delta > 2$.

Now, choose k distinct rational numbers $q_1, q_2, \dots, q_k$ from $\mathbb{Q} \cap [0,1]$. Define the translated sets $Y_i = Y + q_i$.

1. **Disjointness:** As shown previously, these translates are disjoint because Y picks only one element per class.
2. **Containment:** Since $Y \subset [0,1]$ and $q_i \in [0,1]$, it follows that each $Y_i \subset [0,2]$. Thus:

$$\bigcup_{i=1}^{k} Y_i \subset [0,2]$$

3. **The Contradiction:** By monotonicity and normalization, $\mu(\bigcup_{i=1}^{k} Y_i) \leq \mu([0,2]) = 2$. However, by finite additivity and translation invariance:

$$\mu\left(\bigcup_{i=1}^{k} Y_i\right) = \sum_{i=1}^{k} \mu(Y_i) = \sum_{i=1}^{k} \delta = k\delta$$

Since we chose k such that $k\delta > 2$, we get $k\delta \leq 2$, which is a contradiction.

Thus, even finite additivity is inconsistent with translation invariance and normalization.

Remark. *The set Y constructed above is an example of a* ***non-measurable set****. It shows that we cannot assign a meaningful "length" to every possible collection of real numbers if we want to keep the properties of additivity and translation invariance.*

Exercise 1. *Find two disjoint subsets $A, B \subset [0,1]$ such that $\mu(A \cup B) \neq \mu(A) + \mu(B)$.*

Solution. Consider the translates $Y_1, Y_2, \dots, Y_k$ from the construction above. We know that:

$$\mu(Y_1 \cup Y_2 \cup \dots \cup Y_k) \neq \sum_{j=1}^{k} \mu(Y_j)$$

We can find the specific pair by induction:

1. Check Y_1 and Y_2. If $\mu(Y_1 \cup Y_2) \neq \mu(Y_1) + \mu(Y_2)$, we are done. Set $A = Y_1, B = Y_2$.
2. If they are equal, consider the set $(Y_1 \cup Y_2)$ and Y_3. If $\mu((Y_1 \cup Y_2) \cup Y_3) \neq \mu(Y_1 \cup Y_2) + \mu(Y_3)$, then set $A = Y_1 \cup Y_2$ and $B = Y_3$.
3. Continue this process. We must reach a first natural number $r \leq k$ such that:

   $$\mu\left(\left(\bigcup_{j=1}^{r-1} Y_j\right) \cup Y_r\right) \neq \mu\left(\bigcup_{j=1}^{r-1} Y_j\right) + \mu(Y_r)$$

   This stage r is guaranteed to exist because if additivity held for every step up to k, the total sum would equal the measure of the union, which we already proved is impossible.

Taking $A = \bigcup_{j=1}^{r-1} Y_j$ and $B = Y_r$ provides the required disjoint sets. □

σ-Algebras and Measurable Spaces

Definition 1. *Let X be a nonempty set. An* ***algebra*** *of sets on X (also called a* ***field****) is a nonempty collection $\mathcal{A}$ of subsets of X that satisfies the following closure properties:*

1. ***Closed under finite unions:*** *If $E_1, \dots, E_n \in \mathcal{A}$, then $\bigcup_{j=1}^{n} E_j \in \mathcal{A}$.*
2. ***Closed under complements:*** *If $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.*

Definition 2. *A* ***σ-algebra*** *(or* ***σ-field****) is an algebra that is also closed under countable unions. That is, if $\{E_j\}_{j=1}^{\infty} \subset \mathcal{A}$, then $\bigcup_{j=1}^{\infty} E_j \in \mathcal{A}$.*

Remark. *By applying De Morgan's laws, $\bigcap E_j = (\bigcup E_j^c)^c$, we observe that algebras and σ-algebras are also closed under finite and countable intersections, respectively. Furthermore, any algebra $\mathcal{A}$ necessarily contains the empty set $\emptyset$ and the whole set X, since for any $E \in \mathcal{A}$, we have $\emptyset = E \cap E^c$ and $X = E \cup E^c$.*

Proposition. *An algebra $\mathcal{A}$ is a σ-algebra if and only if it is closed under countable disjoint unions.*

Proof. Suppose $\{E_j\}_1^{\infty} \subset \mathcal{A}$. To show the countable union is in $\mathcal{A}$, we can define a sequence of disjoint sets $\{F_k\}$ as follows:

$$F_1 = E_1, \quad F_k = E_k \setminus \left[\bigcup_{j=1}^{k-1} E_j\right] = E_k \cap \left[\bigcup_{j=1}^{k-1} E_j\right]^c \quad \text{for } k > 1$$

These F_k belong to $\mathcal{A}$ because $\mathcal{A}$ is an algebra (closed under finite unions and complements). By construction, the F_k are disjoint and satisfy $\bigcup_1^{\infty} E_j = \bigcup_1^{\infty} F_k$. Thus, if $\mathcal{A}$ is closed under countable disjoint unions, it must also be closed under all countable unions. □

Remark. *The technique of replacing an arbitrary sequence of sets with a disjoint sequence such that their union remains unchanged is a fundamental tool in measure theory.*

Example 1. *Let X be any non-empty set. The following collections of subsets are σ-algebras on X:*

1. ***The Power Set:*** *$\mathcal{P}(X)$, the collection of all subsets of X.*
2. ***The Trivial σ-algebra:*** *$\{\emptyset, X\}$, the smallest possible σ-algebra.*
3. ***The Countable or Co-countable σ-algebra:*** *If X is an uncountable set, then the collection*
$$\mathcal{A} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$
is a σ-algebra.

Remark. *The verification of Example 3 relies on the following logic regarding countable unions:*

- ***Case 1 (All sets are countable):*** *If we have a sequence $\{E_j\}_{j=1}^{\infty} \subset \mathcal{A}$ where every E_j is countable, then their union $\bigcup_{j=1}^{\infty} E_j$ is a countable union of countable sets, which is itself countable. Thus, the union belongs to $\mathcal{A}$.*
- ***Case 2 (At least one set is co-countable):*** *If at least one E_{j_0} is co-countable (meaning $E_{j_0}^c$ is countable), then the complement of the union $(\bigcup E_j)^c = \bigcap E_j^c$ is a subset of $E_{j_0}^c$. Since $E_{j_0}^c$ is countable and any subset of a countable set is countable, the complement of the union is countable. Thus, the union is co-countable and belongs to $\mathcal{A}$.*

Exercise 2. *Prove that the intersection of any family of σ-algebras on a set X is again a σ-algebra on X.*

Solution. Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be a family of σ-algebras on X. Let $\mathcal{M} = \bigcap_{\alpha \in I} \mathcal{M}_\alpha$.

1. Since $X \in \mathcal{M}_\alpha$ for every α, $X \in \mathcal{M}$.
2. If $E \in \mathcal{M}$, then $E \in \mathcal{M}_\alpha$ for all α. Since each $\mathcal{M}_\alpha$ is a σ-algebra, $E^c \in \mathcal{M}_\alpha$ for all α, so $E^c \in \mathcal{M}$.
3. If $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$, then $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}_\alpha$ for all α. Thus $\bigcup E_j \in \mathcal{M}_\alpha$ for all α, implying $\bigcup E_j \in \mathcal{M}$.

□

Definition 3. *Let $\mathcal{E}$ be any subset of the power set $\mathcal{P}(X)$. There exists a unique smallest σ-algebra containing $\mathcal{E}$, denoted by $\mathcal{M}(\mathcal{E})$. It is defined as the intersection of all σ-algebras on X that contain $\mathcal{E}$:*
$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{M} : \mathcal{M} \text{ is a } \sigma\text{-algebra and } \mathcal{E} \subset \mathcal{M}\}$$
*This intersection is well-defined because there is always at least one such σ-algebra, namely $\mathcal{P}(X)$. $\mathcal{M}(\mathcal{E})$ is called the **σ-algebra generated by** $\mathcal{E}$.*

Lemma. *If $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$.*

Proof. By definition, $\mathcal{M}(\mathcal{E})$ is the *smallest* σ-algebra containing the collection of sets $\mathcal{E}$. This means that if $\mathcal{A}$ is any σ-algebra such that $\mathcal{E} \subset \mathcal{A}$, then it must follow that $\mathcal{M}(\mathcal{E}) \subset \mathcal{A}$.

In this case, we are given that $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$. We already know by definition that $\mathcal{M}(\mathcal{F})$ is a σ-algebra. Therefore, $\mathcal{M}(\mathcal{F})$ is a σ-algebra that contains $\mathcal{E}$. By the minimality of $\mathcal{M}(\mathcal{E})$ discussed above, we conclude that $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$. □

Further Definitions in σ-Algebras

Definition 4. *Let $\mathcal{E}$ be a collection of subsets of X. The* ***σ-algebra generated by*** *$\mathcal{E}$, denoted by $\mathcal{M}(\mathcal{E})$, is the unique smallest σ-algebra containing $\mathcal{E}$. It is defined as the intersection of all σ-algebras on X that contain $\mathcal{E}$. Since the power set $\mathcal{P}(X)$ is always one such σ-algebra, this intersection is well-defined.*

Lemma. *If $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$.*

Proof. By definition, $\mathcal{M}(\mathcal{F})$ is a σ-algebra. Since it is given that $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{F})$ is one of the σ-algebras included in the intersection that defines $\mathcal{M}(\mathcal{E})$. Because $\mathcal{M}(\mathcal{E})$ is the intersection of *all* such σ-algebras, it must be a subset of any individual σ-algebra in that collection, specifically $\mathcal{M}(\mathcal{F})$. $\square$

Definition 5. *Let X be a topological space (such as a metric space). The σ-algebra generated by the family of all open sets in X is called the* ***Borel σ-algebra*** *on X, denoted by $\mathcal{B}_X$. The members of $\mathcal{B}_X$ are called* ***Borel sets****.*

Remark. *Since σ-algebras are closed under complements, the Borel σ-algebra is equivalently generated by the family of all closed sets in X. $\mathcal{B}_X$ includes all open sets, closed sets, countable intersections of open sets, and countable unions of closed sets.*

Definition 6. *Within the hierarchy of Borel sets, there is standard terminology for specific levels:*

- *A G_δ* ***set*** *is a countable intersection of open sets.*
- *An F_σ* ***set*** *is a countable union of closed sets.*
- *A $G_{\delta\sigma}$* ***set*** *is a countable union of G_δ sets.*
- *An $F_{\sigma\delta}$* ***set*** *is a countable intersection of F_σ sets.*

The symbols δ and σ originate from the German words Durchschnitt (intersection) and Summe (union).

Generators of the Borel σ-algebra on $\mathbb{R}$

Proposition. *The Borel σ-algebra $\mathcal{B}_\mathbb{R}$ is generated by each of the following collections of intervals:*

a. The open intervals: $\mathcal{E}_1 = \{(a,b) : a < b\}$

b. The closed intervals: $\mathcal{E}_2 = \{[a,b] : a < b\}$

c. The half-open intervals: $\mathcal{E}_3 = \{(a,b] : a < b\}$ or $\mathcal{E}_4 = \{[a,b) : a < b\}$

d. The open rays: $\mathcal{E}_5 = \{(a,\infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty,a) : a \in \mathbb{R}\}$

e. The closed rays: $\mathcal{E}_7 = \{[a,\infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty,a] : a \in \mathbb{R}\}$

Proof. Recall that $\mathcal{B}_\mathbb{R}$ is generated by the collection of all open sets in $\mathbb{R}$. We will use Lemma 1 to show that $\mathcal{M}(\mathcal{E}_j) = \mathcal{B}_\mathbb{R}$ for each j.

Step 1: Establishing $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_\mathbb{R}$ Since every open interval (a,b) is an open set, we have $\mathcal{E}_1 \subset \mathcal{B}_\mathbb{R}$, which implies $\mathcal{M}(\mathcal{E}_1) \subset \mathcal{B}_\mathbb{R}$. Conversely, every open set in $\mathbb{R}$ can be written as a countable union of open intervals. Since σ-algebras are closed under countable unions, any open set must belong to $\mathcal{M}(\mathcal{E}_1)$. Therefore, $\mathcal{B}_\mathbb{R} \subset \mathcal{M}(\mathcal{E}_1)$, and we conclude $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_\mathbb{R}$.

Step 2: Showing $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_\mathbb{R}$ for all j The elements of $\mathcal{E}_j$ for $j \neq 3,4$ are either open or closed sets, which are inherently Borel sets. For $\mathcal{E}_3$ and $\mathcal{E}_4$, notice that:

$$(a,b] = \bigcap_{n=1}^{\infty} \left(a, b + \frac{1}{n}\right) \quad \text{and} \quad [a,b) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b\right)$$

Since these are countable intersections of open sets (G_δ sets), they are Borel sets. Thus, $\mathcal{E}_j \subset \mathcal{B}_\mathbb{R}$ for all j, which implies $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_\mathbb{R}$.

Step 3: Showing $\mathcal{B}_\mathbb{R} \subset \mathcal{M}(\mathcal{E}_j)$ for $j \geq 2$ To prove this, it suffices to show that any open interval (a,b) can be constructed using elements from each $\mathcal{E}_j$.

- **For $\mathcal{E}_2$ (Closed intervals):** We can write $(a,b) = \bigcup_{n=1}^{\infty} [a+\frac{1}{n}, b-\frac{1}{n}]$. Since each $[a+\frac{1}{n}, b-\frac{1}{n}] \in \mathcal{E}_2$, the union is in $\mathcal{M}(\mathcal{E}_2)$.

- **For $\mathcal{E}_3$ (Half-open $(a,b]$):** We can write $(a,b) = \bigcup_{n=1}^{\infty}(a, b-\frac{1}{n}]$. Thus $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_3)$.
- **For $\mathcal{E}_5$ (Open rays (a,∞)):** Note that $(-\infty, b] = (b,\infty)^c$, so $(-\infty,b] \in \mathcal{M}(\mathcal{E}_5)$. Then $(a,b] = (a,\infty) \cap (-\infty, b] \in \mathcal{M}(\mathcal{E}_5)$. Using the result from $\mathcal{E}_3$, we get $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_5)$.
- **For $\mathcal{E}_7$ (Closed rays $[a,\infty)$):** Note that $(a,\infty) = \bigcup_{n=1}^{\infty}[a+\frac{1}{n}, \infty)$. Thus $\mathcal{E}_5 \subset \mathcal{M}(\mathcal{E}_7)$, which implies $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$.

Similar constructions apply to $\mathcal{E}_4, \mathcal{E}_6$, and $\mathcal{E}_8$ by using symmetry and complements. Since $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_j)$ in all cases, we have $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_j)$. □

Product σ-Algebras

Definition 7. *Let $\{X_\alpha\}_{\alpha\in A}$ be an indexed collection of nonempty sets, and let $X = \prod_{\alpha\in A} X_\alpha$ denote their Cartesian product. For each $\alpha \in A$, let $\pi_\alpha : X \to X_\alpha$ be the coordinate maps (or projection maps). If $\mathcal{M}_\alpha$ is a σ-algebra on X_α for each α, the* ***product σ-algebra*** *on X is defined as the σ-algebra generated by the following collection:*

$$\{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{M}_\alpha, \alpha \in A\}$$

This σ-algebra is denoted by $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$.

Remark. *In the specific case where the index set is finite, such as $A = \{1,\dots,n\}$, we use the notation $\bigotimes_1^n \mathcal{M}_j$ or $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$. The importance of this definition is primarily established when discussing the measurability of functions on product spaces.*

Proposition. *If the index set A is countable, then the product σ-algebra $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ is generated by the collection of measurable rectangles:*

$$\mathcal{F} = \left\{ \prod_{\alpha\in A} E_\alpha : E_\alpha \in \mathcal{M}_\alpha \right\}$$

Proof. To prove that $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha = \mathcal{M}(\mathcal{F})$, we show containment in both directions using Lemma 1.

Step 1: Showing $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F})$ Let $\pi_\alpha^{-1}(E_\alpha)$ be a generator of the product σ-algebra, where $E_\alpha \in \mathcal{M}_\alpha$. This inverse image can be written as a special type of product:

$$\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta\in A} E_\beta, \quad \text{where } E_\beta = \begin{cases} E_\alpha & \text{if } \beta = \alpha \\ X_\beta & \text{if } \beta \neq \alpha \end{cases}$$

Since X_β is always in $\mathcal{M}_\beta$, this set is a member of the collection $\mathcal{F}$. Thus, every generator of the product σ-algebra is contained in $\mathcal{F}$. By Lemma 1, it follows that $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F})$.

Step 2: Showing $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ Now consider an arbitrary element of $\mathcal{F}$, which is a product of the form $\prod_{\alpha\in A} E_\alpha$. We can express this product as an intersection of inverse images:

$$\prod_{\alpha\in A} E_\alpha = \bigcap_{\alpha\in A} \pi_\alpha^{-1}(E_\alpha)$$

By the definition of the product σ-algebra, each set $\pi_\alpha^{-1}(E_\alpha)$ is a generator and thus belongs to $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$. Since the index set A is countable, this is a **countable intersection** of measurable sets. Since σ-algebras are closed under countable intersections, the resulting product belongs to $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$.

Therefore, $\mathcal{F} \subset \bigotimes_{\alpha\in A} \mathcal{M}_\alpha$, which by Lemma 1 implies $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha\in A} \mathcal{M}_\alpha$. We conclude that the two σ-algebras are identical. □

Proposition. *Suppose that for each $\alpha \in A$, the σ-algebra $\mathcal{M}_\alpha$ is generated by a collection $\mathcal{E}_\alpha$ (i.e., $\mathcal{M}_\alpha = \mathcal{M}(\mathcal{E}_\alpha)$). Then the product σ-algebra $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ is generated by the collection:*

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}$$

Furthermore, if the index set A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for all α, then $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ is also generated by the collection of "simple" rectangles:

$$\mathcal{F}_2 = \left\{ \prod_{\alpha\in A} E_\alpha : E_\alpha \in \mathcal{E}_\alpha \right\}$$

Proof. We begin by proving the first assertion regarding $\mathcal{F}_1$.

Step 1: Establishing $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha\in A}\mathcal{M}_\alpha$ By definition, each $\mathcal{E}_\alpha \subset \mathcal{M}_\alpha$. Consequently, every set in $\mathcal{F}_1$ is of the form $\pi_\alpha^{-1}(E_\alpha)$ where $E_\alpha \in \mathcal{M}_\alpha$. These are precisely the generators of the product σ-algebra. Since $\mathcal{F}_1 \subset \bigotimes \mathcal{M}_\alpha$, it follows by Lemma 1 that $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes \mathcal{M}_\alpha$.

Step 2: Establishing $\bigotimes_{\alpha\in A}\mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$ For a fixed $\alpha \in A$, consider the collection of subsets of X_α whose inverse images under the coordinate map fall into $\mathcal{M}(\mathcal{F}_1)$:

$$\mathcal{G}_\alpha = \{E \subset X_\alpha : \pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)\}$$

We can verify that $\mathcal{G}_\alpha$ is a σ-algebra on X_α:

- $X_\alpha \in \mathcal{G}_\alpha$ because $\pi_\alpha^{-1}(X_\alpha) = X$, which is in any σ-algebra.
- If $E \in \mathcal{G}_\alpha$, then $\pi_\alpha^{-1}(E^c) = (\pi_\alpha^{-1}(E))^c \in \mathcal{M}(\mathcal{F}_1)$, so $E^c \in \mathcal{G}_\alpha$.
- If $\{E_j\} \subset \mathcal{G}_\alpha$, then $\pi_\alpha^{-1}(\cup E_j) = \cup\pi_\alpha^{-1}(E_j) \in \mathcal{M}(\mathcal{F}_1)$, so $\cup E_j \in \mathcal{G}_\alpha$.

By the construction of $\mathcal{F}_1$, we know that $\mathcal{E}_\alpha \subset \mathcal{G}_\alpha$. Since $\mathcal{M}_\alpha$ is the *smallest* σ-algebra containing $\mathcal{E}_\alpha$, we must have $\mathcal{M}_\alpha \subset \mathcal{G}_\alpha$. This implies that for *any* $E \in \mathcal{M}_\alpha$, the set $\pi_\alpha^{-1}(E)$ belongs to $\mathcal{M}(\mathcal{F}_1)$. Since these inverse images generate the product σ-algebra, we conclude $\bigotimes \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$.

Step 3: Proving the second assertion for $\mathcal{F}_2$ If A is countable and $X_\alpha \in \mathcal{E}_\alpha$, we can use the same logic as the previous Proposition. Every element $\pi_\alpha^{-1}(E_\alpha)$ in $\mathcal{F}_1$ can be written as a product $\prod E_\beta$ where $E_\beta = X_\beta$ for $\beta \neq \alpha$. Since $X_\beta \in \mathcal{E}_\beta$, this product is in $\mathcal{F}_2$. Thus $\mathcal{F}_1 \subset \mathcal{F}_2$, and conversely, every element of $\mathcal{F}_2$ is a countable intersection of elements from $\mathcal{F}_1$:

$$\prod_{\alpha\in A} E_\alpha = \bigcap_{\alpha\in A} \pi_\alpha^{-1}(E_\alpha)$$

The result then follows directly from the closure of σ-algebras under countable intersections. $\square$